



DATA SCIENCE &  
ARTIFICIAL INTELLIGENCE

# Probabilistic Machine Learning

## Bayesian classification

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# Linear classification

$D = (x_n, y_n)$  with  $n = 1, \dots, N$

binary  $y_n \in \{0, 1\}$ .

multi class  $y_n = (y_{nj})_{j=1, \dots, K}$

**Discriminative approach**  $p(C_k|x) = f(h(x))$

$$p(C_k|x) = f(w^T \phi(x))$$

logit  $\sigma(a) = \frac{1}{1 + e^{-a}}$

probit  $\psi(a) = \int_{-\infty}^a \mathcal{N}(\theta|0, 1) d\theta$

# Logistic regression (binary)

$$p(\underline{y}|\underline{x}) = \prod_{n=1}^N s_n^{y_n} (1 - s_n)^{1-y_n}$$

$$E(w) = -\frac{1}{N} \log p(\underline{y}|\underline{x}) = -\frac{1}{N} \sum_{n=1}^N y_n \log s_n + (1 - y_n) \log(1 - s_n)$$

$$\nabla E(w) = \frac{1}{N} \sum_{n=1}^N (s_n - y_n) \phi(x_n)$$

$$w_{n+1} = w_n - \eta_n \nabla E(w_n)$$

# Logistic regression (multi class)

$$p(C_k|x) = \sigma_k(W^T \phi(x))$$

$$\text{softmax} \quad \sigma_k(\underline{a}) = \frac{\exp(a_k)}{\sum_j \exp(a_j)} \quad \begin{cases} a_1 = w_1^T \phi(x) \\ \dots \\ a_k = w_k^T \phi(x) \end{cases}$$

$$p(C_k|x) = \sigma_k(W^T \phi(x)) = \sigma_k(a_1, \dots, a_k)$$

$$E(w) = -\frac{1}{N} \sum_{n=1}^N \sum_{j=1}^K y_{nj} \log s_{nj} \quad s_{nj} = \sigma_j(W^T \phi(x_n)).$$

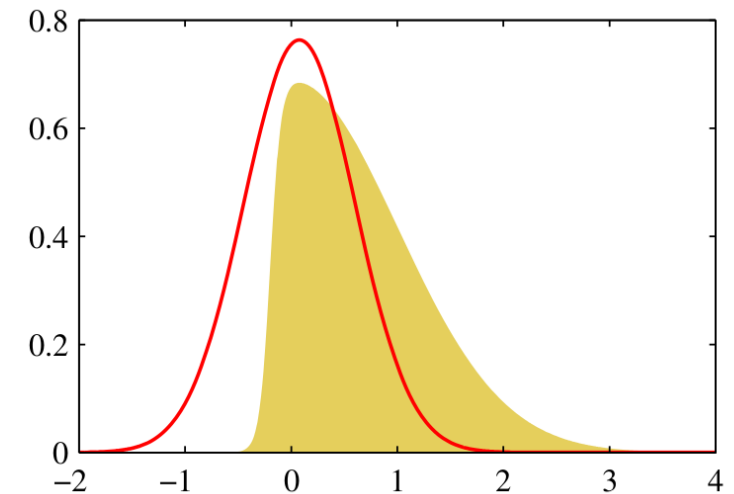
$$\nabla_{w_j} E(w) = \frac{1}{N} \sum_{n=1}^N (s_{nj} - y_{nj}) \phi(x_n)$$



# Laplace Approximation

$$p(z) = \frac{1}{Z} f(z) \quad Z = \int f(z) dz$$

- find a mode  $z_0$  of  $f(z)$ , i.e. a point such that  $\frac{d}{dz} f(z_0) = 0$  and  $z_0$  is a point of maximum;
- match the curvature of  $f$  at  $z_0$  with a normal distribution.



$$\log f(z) \approx \log f(z_0) - \frac{1}{2} A (z - z_0)^2$$

$$A = -\frac{d^2}{dz^2} \log f(z_0), A > 0$$

$$f(z) \approx f(z_0) \cdot \exp\left(-\frac{1}{2} A (z - z_0)^2\right)$$

$$q(z) \sim \mathcal{N}(z|z_0, A^{-1})$$

$$q(z) = \left(\frac{A}{2\pi}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2} A (z - z_0)^2\right)$$

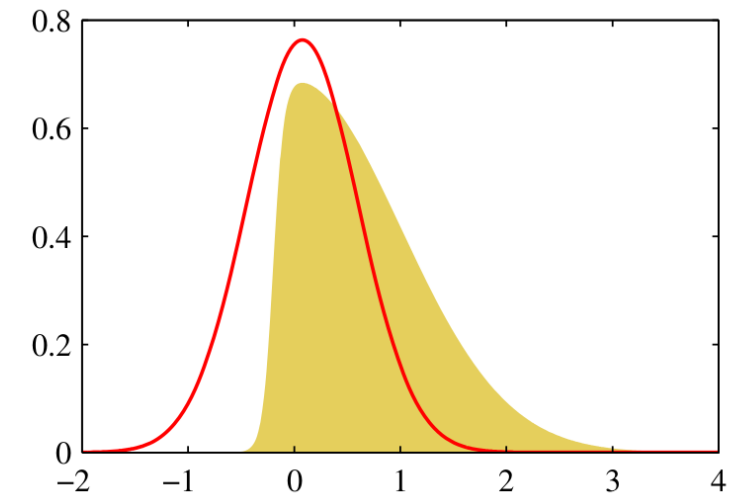
$$Z = \frac{(2\pi)^{\frac{1}{2}}}{|A|^{\frac{1}{2}}} f(z_0)$$

# Laplace Approximation (n dim)

$$\log f(z) \approx \log f(z_0) - \frac{1}{2}(z - z_0)^T A (z - z_0)$$

$$A = -\nabla \nabla \log f(z_0).$$

$$q(z) = \mathcal{N}(z|z_0, A^{-1}) \quad Z = \frac{(2\pi)^{\frac{k}{2}}}{|A|^{\frac{1}{2}}} f(z_0)$$



# Laplace for model comparison

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}. \quad f(\theta) = p(D|\theta)p(\theta) \quad Z = p(D)$$

Laplace approximation around the maximum-a-posteriori (MAP)

$$p(D) \approx \frac{(2\pi)^{\frac{M}{2}}}{|A|^{\frac{1}{2}}} f(\theta_{MAP}) = \frac{(2\pi)^{\frac{M}{2}}}{|A|^{\frac{1}{2}}} p(D|\theta_{MAP})p(\theta_{MAP})$$

$$\log p(D) \approx \log p(D|\theta_{MAP}) + \log p(\theta_{MAP}) + \frac{M}{2} \log(2\pi) - \frac{1}{2} \log |A| \quad M = |\theta|$$

## Bayesian Information Content (BIC)

$$\log p(D) \approx \log p(D|\theta_{MAP}) - \frac{1}{2} M \log N$$

# Bayesian Logistic Regression

$(x_n, y_n)$  with  $n = 1, \dots, N$ ,

$\sigma(w^T \phi(x))$ .    logit

$$p(\underline{y}|w, \underline{x}) = \prod_{i=1}^N s_i^{y_i} (1 - s_i)^{1-y_i}$$

log posterior

$$\log p(w|\underline{y}, \underline{x}) = \log p(w) + \log p(\underline{y}|w) + C$$

$$= -\frac{1}{2}(w - m_0)^T S_0^{-1}(w - m_0) +$$

$$+ \sum_{i=1}^N [y_i \log s_i(w) + (1 - y_i) \log(1 - s_i(w))] + C$$

# Bayesian Logistic Regression

## Laplace approximation

1. find  $w_{MAP} = \operatorname{argmax}_w \log p(w|\underline{y}) = \operatorname{argmax}_w \log p(w) + \log p(\underline{y}|w)$

2.  $S_N^{-1} = S_0^{-1} + \sum_{n=1}^N s_n(w_{MAP})(1 - s_n(w_{MAP}))\phi(x_n)\phi^T(x_n)$

$$q(w) = \mathcal{N}(w|w_{MAP}, S_N)$$

## predictive

$$p(C_1|x^\star, \underline{y}, \underline{x}) = \int p(C_1|x^\star, w, \underline{x}, \underline{y})p(w|\underline{y}, \underline{x})dw \approx \int \sigma(w^T \phi(x^\star))q(w)dw$$

$$a = w^T \phi(x^\star), \quad q(a) \sim \mathcal{N}(a|\mu_a, \sigma_a^2), \quad \mu_a = w_{MAP}^T \phi(x^\star), \quad \sigma_a^2 = \phi^T(x^\star)S_N\phi(x^\star)$$

$$p(C_1|x^\star, \underline{y}, \underline{x}) = \int \sigma(a)q(a)da$$

# Bayesian Logistic Regression

$$p(C_1|x^\star, \underline{y}, \underline{x}) = \int \sigma(a)q(a)da$$

$$\sigma(a) \approx \psi(\lambda a), \text{ such that } \lambda^2 = \frac{\pi}{8} \text{ and } \sigma'(0) = \psi'(0)$$

$$\int \psi(\lambda a)q(a)da = \int \psi(\lambda a)\mathcal{N}(a|\mu_a, \sigma_a^2)da = \Psi\left(\frac{\mu_a}{(\lambda^{-2} + \sigma_a^2)^{\frac{1}{2}}}\right)$$

$$p(C_1|x^\star, \underline{y}, \underline{x}) \approx \sigma(\kappa(\sigma_a^2)\mu_a) \quad \kappa(\sigma_a^2) = \left(1 + \pi \frac{\sigma_a^2}{8}\right)^{-\frac{1}{2}}$$

- ▶ if  $\sigma_a^2 = 0$  then  $p(C_1|x^\star, \underline{y}, \underline{x}) \approx \sigma(\mu_a)$
- ▶ if  $\sigma_a^2 \gg 0$  then  $p(C_1|x^\star, \underline{y}, \underline{x}) \rightarrow \frac{1}{2}$  which represents the maximum level of uncertainty on the prediction

# Bayesian Logistic Regression

